STAT210 Statistical Methods I: Linear Model

Lec 12: Multiple Linear Regression - Categorical Variables

Zhaoxia Yu
Professor, Department of Statistics

2026-11-09

```
library(tidyr) #the pipe (%>%) tool is extremely useful
library(MASS)
library(ggplot2)
library(reshape2)
library(ggcorrplot)
library(car)
```

Section 1

Review

Review: Lecture 01-06

- Lectures 01-06: SLR: least squares estimate, inference (CI, prediction, t-test, F-test), diagnostics, transformations, ANOVA, lack of fit test.
- Lectures 07-11: Multiple Linear Regression (MLR).
- Lecture 11: multicollinearity
- This lecture focuses on categorical predictor variables.

Section 2

Categorical Predictors

Continuous vs Categorical Predictor Variables

- So, in both SLR and MLR, we have focused on continuous predictor variables (also called quantitative predictor variables).
  - E.g., in `mtcars`, the variable 'wt' (weight) is a continuous predictor variable.
  - E.g., in 'cdi' (northeast region), the variable 'hospitals_std' is a continuous predictor variable.
- However, in practice, we often have categorical predictor variables as well.
  - E.g., in `mtcars`, the variable 'vs' (engine type) is a categorical predictor variable.
  - E.g., in 'cdi' (northeast region), the variable 'state' is a categorical predictor variable.

Definition of Categorical Predictor Variables

- Categorical predictor variables (also called qualitative predictor variables or factors) take on a limited number of values, representing different groups or categories.
- E.g., in `mtcars`,
  - the variable 'vs' indicates whether the engine is V-shaped (0) or straight (1);
  - the variable 'am' indicates whether the transmission is automatic (0) or manual (1).

Categorical Predictor Variables with More than Two Categories

- E.g., in 'cdi' (northeast region), the state variable has the following categories:
  - CT: Connecticut
  - DE: Delaware
  - MA: Massachusetts
  - ME: Maine
  - NH: New Hampshire
  - NJ: New Jersey
  - NY: New York
  - PA: Pennsylvania
  - RI: Rhode Island
  - VT: Vermont

Section 3

Mathematical Representation

Convert Categorical Predictor Variables to Numerical Values

- We will present mathematical representation of categorical predictor variables.
  - This is automatically done in R and other software.
  - We present it to better understand the inner workings of the regression model with categorical predictors.
- In SLR or MLR, the design matrix X contains numerical values.
- Indicator variables will be created for a categorical predictor variable to represent the different categories numerically.
- Example, PlantGrowth dataset:
  - group: a categorical predictor variable with three categories: ctrl, trt1, trt2.

Convert Categorical Predictor Variables to Numerical Values

- The group variable is a categorical variable with three categories: ctrl, trt1, trt2.
- We can create three indicator variables to represent the three categories:
  - X_{ctrl}: 1 if group=ctrl, 0 otherwise
  - X_{trt1}: 1 if group=trt1, 0 otherwise
  - X_{trt2}: 1 if group=trt2, 0 otherwise
- The design matrix X will then include these three indicator variables along with the intercept term.

Convert Categorical Predictor Variables to Numerical Values

- PlantGrowth

```
head(PlantGrowth)
```

```
##   weight group
## 1   4.17  ctrl
## 2   5.58  ctrl
## 3   5.18  ctrl
## 4   6.11  ctrl
## 5   4.50  ctrl
## 6   4.61  ctrl
```

```
summary(PlantGrowth)
```

```
##      weight       group
##  Min.   :3.590   ctrl:10
##  1st Qu.:4.550   trt1:10
##  Median :5.155   trt2:10
##  Mean   :5.073
##  3rd Qu.:5.530
##  Max.   :6.310
```

Convert Categorical Predictor Variables to Numerical Values

- Create indicator variables in R:

```
# creating three indicator variables
ctrl <- ifelse(PlantGrowth$group=="ctrl", 1, 0)
trt1 <- ifelse(PlantGrowth$group=="trt1", 1, 0)
trt2 <- ifelse(PlantGrowth$group=="trt2", 1, 0)
design_matrix <- data.frame(Intercept=1, ctrl=ctrl, trt1=trt1, trt2=trt2)
head(design_matrix)
```

```
##   Intercept ctrl trt1 trt2
## 1         1    1    0    0
## 2         1    1    0    0
## 3         1    1    0    0
## 4         1    1    0    0
## 5         1    1    0    0
## 6         1    1    0    0
```

```
summary(design_matrix)
```

```
##    Intercept      ctrl            trt1            trt2
##  Min.   :1   Min.   :0.0000   Min.   :0.0000   Min.   :0.0000
##  1st Qu.:1   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.:0.0000
##  Median :1   Median :0.0000   Median :0.0000   Median :0.0000
##  Mean   :1   Mean   :0.3333   Mean   :0.3333   Mean   :0.3333
##  3rd Qu.:1   3rd Qu.:1.0000   3rd Qu.:1.0000   3rd Qu.:1.0000
##  Max.   :1   Max.   :1.0000   Max.   :1.0000   Max.   :1.0000
```

Convert Categorical Predictor Variables to Numerical Values

- In this example, we have three indicator variables for the three categories.
- To compute the LSEs, we would need to invert the matrix $X^T X$, where X is the design matrix including these three indicator variables.

```
X <- as.matrix(design_matrix)
XTX <- t(X) %*% X
XTX
```

```
##           Intercept ctrl trt1 trt2
## Intercept        30   10   10   10
## ctrl             10   10    0    0
## trt1             10    0   10    0
## trt2             10    0    0   10
```

Convert Categorical Predictor Variables to Numerical Values

- The matrix $X^T X$ is singular and cannot be inverted.
- This is because there is linear dependency in the design matrix:

```
ctrl + trt1 + trt2
```

```
##  [1] 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
```

- The sum of last three columns equals the first column.
- To avoid this issue, we need to use only two indicator variables for the three categories.

Convert Categorical Predictor Variables to Numerical Values

- We can use only two indicator variables for the three categories:
  - X_{ctrl}: 1 if group=ctrl, 0 otherwise
  - X_{trt1}: 1 if group=trt1, 0 otherwise
- The third category (trt2) is represented by both X_{ctrl}=0 and X_{trt1}=0.
- The design matrix X will then include these two indicator variables along with the intercept term.

Convert Categorical Predictor Variables to Numerical Values

- The design matrix X now includes only two indicator variables for the three categories.

```
design_matrix <- data.frame(Intercept=1, ctrl=ctrl, trt1=trt1)
X <- as.matrix(design_matrix)
XTX <- t(X) %*% X
beta_hat <- solve(XTX) %*% t(X) %*% as.matrix(PlantGrowth$weight)
t(beta_hat)
```

```
##      Intercept   ctrl   trt1
## [1,]     5.526 -0.494 -0.865
```

Dummy Variables

- The two indicator variables created above are called dummy variables.
- In general, for a categorical predictor variable with k categories, we need to create $k - 1$ dummy variables to represent the categories in the design matrix.
- The category that is not represented by a dummy variable is called the reference category.

Compare with R's lm function

```
lm_fit <- lm(weight ~ group, data=PlantGrowth)
summary(lm_fit)$coefficients[,1]
```

```
## (Intercept)   grouptrt1   grouptrt2
##       5.032      -0.371       0.494
```

- Do they give the same results?
- No. Why?

Section 4

Reference Category

The Default Reference Category in R

- By default, R uses the first category (in alphabetical order) as the reference category.
- In the PlantGrowth dataset, the categories are 'ctrl', 'trt1', and 'trt2'. The reference category is 'ctrl'.
- The dummy variables created by R's `lm` function are:
  - X_{trt1}: 1 if group=trt1, 0 otherwise
  - X_{trt2}: 1 if group=trt2, 0 otherwise
  - The design matrix X used by R's `lm` function includes these two dummy variables along with the intercept term.

The Default Reference Category in R

```
design_matrix <- data.frame(Intercept=1, trt1=trt1, trt2=trt2)
X <- as.matrix(design_matrix)
XTX <- t(X) %*% X
beta_hat <- solve(XTX) %*% t(X) %*% as.matrix(PlantGrowth$weight)
t(beta_hat)
```

```
##      Intercept   trt1  trt2
## [1,]     5.032 -0.371 0.494
```

Change the Reference Category

- We can change the reference category by releveling the factor variable using the `relevel` function in R.
- For example, if we have a good reason to use trt1 as the reference category, we can do the following:

```
PlantGrowth$group_relevel <- relevel(PlantGrowth$group, ref="trt1")
lm_fit_relevel <- lm(weight ~ group_relevel, data=PlantGrowth)
summary(lm_fit_relevel)$coefficients[,1]
```

```
##       (Intercept) group_relevelctrl group_releveltrt2
##             4.661             0.371             0.865
```

Interpretation of Coefficients with Categorical Predictors

- The interpretation of the coefficients in the presence of categorical predictor variables depends on the reference category.
- For example, in the PlantGrowth dataset with 'ctrl' as the reference category:
  - The intercept represents the mean weight for the 'ctrl' group.
  - The coefficient for 'trt1' represents the difference in mean weight between the 'trt1' group and the 'ctrl' group.
  - The coefficient for 'trt2' represents the difference in mean weight between the 'trt2' group and the 'ctrl' group.

Interpretation of Coefficients with Categorical Predictors

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)   5.0320     0.1971  25.527   <2e-16 ***
grouptrt1    -0.3710     0.2788  -1.331   0.1944
grouptrt2     0.4940     0.2788   1.772   0.0877 .
```

- Here, 5.03 is the sample mean of the 'ctrl' group; -0.37 is difference in sample means between 'trt1' and 'ctrl'; 0.49 is difference between 'trt2' and 'ctrl'.
- How about the p-values?
  - The p-value for 'trt1' tests the null hypothesis that the mean weight of 'trt1' is equal to that of 'ctrl'.

Interpretation of Coefficients with Categorical Predictors

	Estimate	Std. Error	t value	Pr(>|t|)
(Intercept)	4.661	0.1971284	23.644492	1.405827e-19
group_relevelctrl	0.371	0.2787816	1.330791	1.943879e-01
group_releveltrt2	0.865	0.2787816	3.102787	4.459236e-03

- The intercept 4.66 is the sample mean of the 'trt1' group; 0.37 is difference in sample means between 'ctrl' and 'trt1'; 0.87 is difference between 'trt2' and 'trt1'.
- How about the p-values?
  - The p-value for 'ctrl' tests the null hypothesis that the mean weight of 'ctrl' is equal to that of 'trt1'.

Section 5

Alternatives to Reference Category

Alternatives to Reference Category

- Instead of choosing a reference category, we can also
  - include all categories in the model and remove the intercept term, or
  - use effect coding (also called deviation coding) to represent categorical predictor variables.

No Intercept Model

- We can include all categories in the model and remove the intercept term.
- Example, PlantGrowth dataset without intercept:

```
design_matrix_no_intercept <- data.frame(ctrl=ctrl, trt1=trt1, trt2=trt2)
head(design_matrix_no_intercept)
```

```
##   ctrl trt1 trt2
## 1    1    0    0
## 2    1    0    0
## 3    1    0    0
## 4    1    0    0
## 5    1    0    0
## 6    1    0    0
```

```
X_no_intercept <- as.matrix(design_matrix_no_intercept)
XTX_no_intercept <- t(X_no_intercept) %*% X_no_intercept
beta_hat_no_intercept <- solve(XTX_no_intercept) %*% t(X_no_intercept) %*% as.matrix(PlantGrowth$weight)
beta_hat_no_intercept
```

```
##       [,1]
## ctrl 5.032
## trt1 4.661
## trt2 5.526
```

No Intercept Model

- In R, we can fit the model without intercept using the `lm` function:

```
lm_fit_no_intercept <- lm(weight ~ group + 0, data=PlantGrowth)
# or lm_fit_no_intercept <- lm(weight ~ group - 1, data=PlantGrowth)
summary(lm_fit_no_intercept)$coefficients
```

```
##           Estimate Std. Error  t value     Pr(>|t|)
## groupctrl    5.032  0.1971284 25.52651 1.936575e-20
## grouptrt1    4.661  0.1971284 23.64449 1.405827e-19
## grouptrt2    5.526  0.1971284 28.03250 1.688656e-21
```

Effect Coding

- In effect coding, we create k indicator variables for a categorical predictor variable with k categories.
- The indicator variables are coded as follows:
  - For category 1: X_1=1, X_2=0, ..., X_k=0
  - For category 2: X_1=0, X_2=1, ..., X_k=0
  - ...
  - For category k-1: X_1=0, X_2=0, ..., X_{k-1}=1
  - For category k: X_1=-1, X_2=-1, ..., X_{k-1}=-1

Effect Coding

- The design matrix X will then include these k indicator variables along with the intercept term.
- Example, PlantGrowth dataset with effect coding:

```
ctrl_ec <- ifelse(PlantGrowth$group=="ctrl", 1, ifelse(PlantGrowth$group=="trt1", 0, -1))
trt1_ec <- ifelse(PlantGrowth$group=="trt1", 1, ifelse(PlantGrowth$group=="ctrl", 0, -1))
design_matrix_ec <- data.frame(Intercept=1, ctrl_ec=ctrl_ec, trt1_ec=trt1_ec)
X_ec <- as.matrix(design_matrix_ec)
XTX_ec <- t(X_ec) %*% X_ec
beta_hat_ec <- solve(XTX_ec) %*% t(X_ec) %*% as.matrix(PlantGrowth$weight)
t(beta_hat_ec)
```

```
##      Intercept ctrl_ec trt1_ec
## [1,]     5.073  -0.041  -0.412
```

Effect Coding

```
t(beta_hat_ec)
     Intercept ctrl_ec trt1_ec
[1,]     5.073  -0.041  -0.412‘‘‘
```

- How to interpret the coefficients here?
  - 5.07 is the overall mean weight across all groups.
  - -0.04 is the deviation of the ‘ctrl’ group mean from the overall mean.
  - -0.41 is the deviation of the ‘trt1’ group mean from the overall mean.
  - The deviation of the ‘trt2’ group mean from the overall mean can be calculated as -(-0.04 + -0.41) = 0.45.

Effect Coding

- Let's verify the calculations:

```
mean_ctrl <- mean(PlantGrowth$weight[PlantGrowth$group=="ctrl"])
mean_trt1 <- mean(PlantGrowth$weight[PlantGrowth$group=="trt1"])
mean_trt2 <- mean(PlantGrowth$weight[PlantGrowth$group=="trt2"])
overall_mean <- mean(PlantGrowth$weight)
deviation_ctrl <- mean_ctrl - overall_mean
deviation_trt1 <- mean_trt1 - overall_mean
deviation_trt2 <- mean_trt2 - overall_mean
deviation_ctrl
```

```
## [1] -0.041
```

```
deviation_trt1
```

```
## [1] -0.412
```

```
deviation_trt2
```

```
## [1] 0.453
```

Effect Coding

- In R, we can use the `contr.sum` function to specify effect coding for a factor variable.

```
PlantGrowth$group_ec <- factor(PlantGrowth$group,
                               levels=c("ctrl", "trt1", "trt2"),
                               labels=c("ctrl", "trt1", "trt2"))
contrasts(PlantGrowth$group_ec) <- contr.sum(3)
lm_fit_ec <- lm(weight ~ group_ec, data=PlantGrowth)
summary(lm_fit_ec)$coefficients
```

```
##             Estimate Std. Error    t value     Pr(>|t|)
## (Intercept)    5.073  0.1138121 44.5734621 8.020008e-27
## group_ec1     -0.041  0.1609546 -0.2547302 8.008617e-01
## group_ec2     -0.412  0.1609546 -2.5597274 1.639014e-02
```

- The coefficients represent the deviation of each category's mean from the overall mean.

The contrasts Function

- The `contrasts` function in R allows us to specify how categorical predictor variables are coded in the design matrix.
- By default, R uses treatment contrasts (reference category coding).
- We can use `contr.sum` for effect coding or `contr.treatment` to specify a different reference category.
- The 'contr.treatment' option allows us to specify which category to use as the reference category.

```
contrasts(PlantGrowth$group) <- contr.treatment(3, base=2) # use 'trt1' as reference
lm_fit_contr <- lm(weight ~ group, data=PlantGrowth)
summary(lm_fit_contr)$coefficients
```

```
##             Estimate Std. Error   t value     Pr(>|t|)
## (Intercept)    4.661  0.1971284 23.644492 1.405827e-19
## group1         0.371  0.2787816  1.330791 1.943879e-01
## group3         0.865  0.2787816  3.102787 4.459236e-03
```

- The coefficients now represent the differences in means between each category and the 'trt1' group.

The contrasts Function

- It is good habit to always check the contrasts used for categorical predictor variables in your model.
- After fitting the model, you can check the contrasts used by:

```
contrasts(PlantGrowth$group)
```

```
##      1 3
## ctrl 1 0
## trt1 0 0
## trt2 0 1
```

- It is also a good habit to set to the default contrasts after fitting the model to avoid confusion in future analyses.

```
contrasts(PlantGrowth$group) <- contr.treatment(3)
```

Section 6

Overall Effect

The Overall Effect of a Categorical Predictor Variable

- We might be interested in knowing whether there is an overall effect of a categorical predictor variable on the response variable.
- For example, in the PlantGrowth dataset, we might want to test whether there is any difference in mean weight among the three groups (ctrl, trt1, trt2).
  - H_0: $\mu_{ctrl} = \mu_{trt1} = \mu_{trt2}$
  - H_a: At least one group mean is different from another one
- Which test should we use?

The Overall Effect of a Categorical Predictor Variable

- We can use the ANOVA F-test to test the overall effect of a categorical predictor variable.
- We can find the result from anova of the summary of the lm fit.

```
anova(lm_fit)
```

```
## Analysis of Variance Table
##
## Response: weight
##           Df  Sum Sq Mean Sq F value  Pr(>F)
## group      2  3.7663  1.8832  4.8461 0.01591 *
## Residuals 27 10.4921  0.3886
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
```

The Overall Effect of a Categorical Predictor Variable

```
summary(lm_fit)
```

```
##
## Call:
## lm(formula = weight ~ group, data = PlantGrowth)
##
## Residuals:
##     Min      1Q  Median      3Q     Max
## -1.0710 -0.4180 -0.0060  0.2627  1.3690
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   5.0320     0.1971  25.527   <2e-16 ***
## grouptrt1    -0.3710     0.2788  -1.331   0.1944
## grouptrt2     0.4940     0.2788   1.772   0.0877 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.6234 on 27 degrees of freedom
## Multiple R-squared:  0.2641, Adjusted R-squared:  0.2096
## F-statistic: 4.846 on 2 and 27 DF,  p-value: 0.01591
```

Section 7

Categorical Variables in MLR

Categorical Predictor Variables in MLR

- For a categorical predictor with more than two categories, the model is already a multiple linear regression model because it includes multiple dummy variables.
- A MLR can include both continuous and categorical predictor variables.
  - E.g., in `mtcars`, we can fit a MLR model with 'mpg' as the response variable, 'wt' (continuous) and 'vs' (categorical) as predictor variables.

```
lm_mlr <- lm(mpg ~ wt + vs, data=mtcars)
```

Categorical and Continuous Predictor Variables in MLR

```
summary(lm_mlr)$coefficients
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  33.0042     2.3554  14.012 1.92e-14 ***
wt           -4.4428     0.6134  -7.243 5.63e-08 ***
vs            3.1544     1.1907   2.649   0.0129 *
```

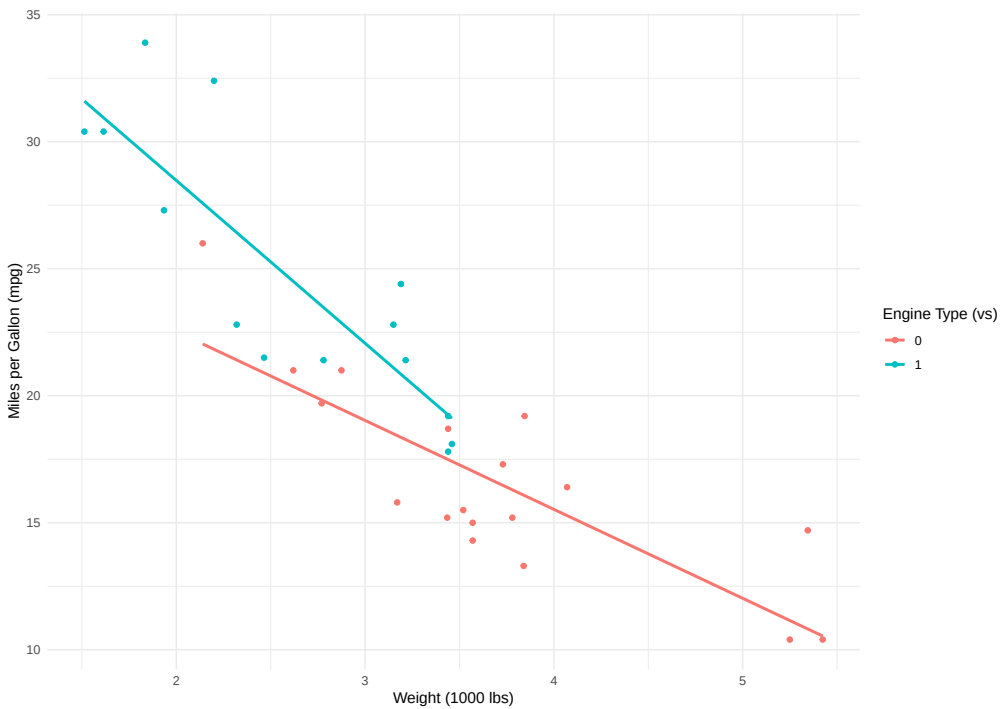
- How to interpret the coefficients here?
  - 33.00 the estimated mean mpg when wt=0 and vs=0 (V-shaped engine).
  - -4.44 represents the change in mean mpg for a one-unit increase in wt, holding vs constant (within the same engine type).
  - 3.15 represents the difference in mean mpg between straight engine (vs=1) and V-shaped engine (vs=0), holding wt constant.

Interactions between Categorical and Continuous Predictors

- What assumptions did we make about the meaningfulness of these interpretations?
  - We assume that the effect of wt on mpg is the ***same*** for both engine types (no interaction effect).
- If we suspect that the effect of wt on mpg differs by engine type, we can include an interaction term between wt and vs in the model.
- It is always helpful to visualize the data to check for potential interaction effects.

Visual Inspection of Interaction Effects

```
ggplot(mtcars, aes(x=wt, y=mpg, color=factor(vs))) +
  geom_point() +
  geom_smooth(method="lm", se=FALSE) +
  labs(color="Engine Type (vs)", x="Weight (1000 lbs)", y="Miles per Gallon (mpg)") +
  theme_minimal()
```



- This will be in next lecture.

Section 8

Conclusion

Conclusion

- We showed how do categorical predictor variables work in linear regression models.
- We showed the mathematical representation of categorical predictor variables using dummy variables.
- The mathematical representation helps us understand how the regression model handles categorical predictors; it shows the inner workings of the model.
- In R or other software, the process is done automatically when fitting the model.

Conclusion

- The different ways to represent categorical predictors (reference category, no intercept, effect coding) lead to different interpretations of the coefficients, but they all model the same underlying relationships in the data.
- We shown two methods to change the default reference category in R: `relevel` function and `contrasts` function.
- For the GrowthPlants example, what is your favorite way to represent the categorical predictor variable 'group'?
- Lastly, the overall effect of a categorical predictor variable can be tested using the ANOVA F-test.